Supplementary material: A model-agnostic dual-path supervisory logic for transformer differential protection, evaluated with a leak-free benchmark of six reproduced methods

S1 Full tables and notes

Tables A–D of the paper with their complete notes (search budget, appendix variants of the reproductions, all post-hoc labels, per-seed instability flags, common-set sizes), and the operate-time table including the like-for-like rows. The cells are identical to the manuscript tables; they are generated by `tools/publication_tables_AD.py` and `tools/paper_operate_times.py` from the saved per-trace files.

Table S1: Six published methods re-implemented and trained on the benchmark data (native decision logic and like-for-like mode), against the proposed relay. Dependability / security in %; CV = plant A, 3 seeds × 5 folds, common set (3,302 traces), mean over seeds; OOD and plant B: one final model scored once (plant B zero-shot, logic and threshold frozen from plant A). The reproductions are re-implementations trained on our data, not the authors' code.

Method	Mode	CV		OOD		Plant B		Rate	Latency per decision (median / budget)
		dep.	sec.	dep.	sec.	dep.	sec.		
Simões et al. 2021 (RT-BSWT + 2 SVMs; C, γ searched)	native	69.7	92.2	70.9	73.8	83.2	89.4	12.8 kHz	7.5 / 78.1 µs per sample; SVM 1 498 µs per detection
	like-for-like	76.8	72.1	75.0	60.4	86.7	71.7		
Ngaopitakkul & Kunakorn 2006 (db4 DWT + BPNN)$^{\triangle}$	native‡	27.7	99.3	20.3	99.3	28.0	97.6	20 kHz	3.3 / 50 µs per sample
	like-for-like	38.4	97.5	35.1	98.0	61.3	90.5		
Abdusalam et al. 2022 (DWT + OMP + BA + SVM)	native	76.4	99.9	83.1	94.0	71.4	99.7	10 kHz	609 / 2.5 ms per window: not real-time
	like-for-like	79.2	92.9	84.5	86.6	89.2	95.5		
Afrasiabi et al. 2020 (CNN, with Hajipour CT stage)	native‡	86.7	88.6	95.3	94.6	87.5	75.2	5.6 kHz	117 / 1250 µs per decision
	like-for-like	87.3	99.6	86.5	100.0	93.9	99.9		
Key et al. 2025 (Com-DNN + Dis-DNN)	native‡	97.0	75.3	96.6	78.5	98.2	69.7	3.2 kHz	37.0 / 312.5 µs per sample
	like-for-like	97.2	99.6	98.0	96.0	96.9	98.9		
Alhamd et al. 2024 (RT-BSWT + BiLSTM)	native‡	96.5	82.9	97.3	87.2	92.6	83.8	3 kHz	1166–1169 / 1667 µs per decision (trained configs; largest search-space config 1799 µs)
	like-for-like	97.9	98.2	95.3	95.3	90.0	96.2		
Proposed relay (deployed LSTM, dual-path)	hybrid	93.0	99.0	98.0	96.0	92.7	88.4	1.6 kHz	140 / 625 µs (ONNX)
LSTM alone (same logic, no 87T path)	standalone	97.7	95.6	99.3	87.3	92.9	87.3		

Notes. Like-for-like = 20-sample dwell (12.5 ms) with a threshold tuned on validation traces (plateau rule) plus the relay's high-set element; Simões and Ngaopitakkul act once per detector firing (no dwell). Native = the method's own decision rule (the "as intended" row). Reproductions are scored on the traces that exist in the 20 kHz re-simulation (plant A 3,302 of 3,332, OOD 297, plant B 1,278); the proposed-relay rows use all evaluable traces (OOD 299, plant B 1,257). ‡ fold-model standard deviation > 3 pt: report the range, do not rank by the mean. $^{\triangle}$ the paper's sanity check holds on noise-free currents but not on the benchmark's CT secondaries with measurement noise (detector fires in 12–38 % of internal-fault traces; spec amendment 8); the lumped plants have almost no energy above 1.6 kHz (no winding capacitances), which also limits Simões. Variants in the appendix: Simões with the published C, γ 59.9 / 95.3; Afrasiabi without the Hajipour stage 86.4 / 88.1 (the stage is nearly inactive on our CT-stress traces); Abdusalam GA + SVM 77.2 / 99.9; Alhamd with the reduced CPU budget 85.8 / 54.9 (native, CV). Simões' detector factor was chosen on validation traces (50 or 100): it finds 86–92 % of events, which caps its dependability (decision rule fixed in a dated plan before the analysis). Latency: single thread, development PC, Python; networks at the largest search-space size except Alhamd (trained configurations listed); n/e = not evaluated. Paired comparisons with the proposed relay (native and like-for-like, 48 tests) are listed in Supplementary Table S3. Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 48 tests. Rows use the second (full) run of each reproduction; Key and Afrasiabi training is not bit-reproducible (Supplementary Section S2).

Table S2: Same 12 DWT + second-harmonic features, different classifiers: default configuration (B1: equal training routine) and tuned with an equal 12-configuration search (B2); standalone and inside the dual-path logic (hybrid). Dependability / security in %, plant-A CV, 3 seeds × 5 folds, common set, mean over seeds.

Classifier (architecture)	Default configuration				Tuned, equal budget				unique configs of 12	chosen configuration
	standalone		hybrid		standalone		hybrid			
	dep.	sec.	dep.	sec.	dep.	sec.	dep.	sec.		
SVM (pooled features)	84.9	75.1	81.7	96.9	90.9	90.4	87.0	97.7	10	n components 240, gamma x 1.0, alpha 1e-05
Random forest (pooled features)	96.8	95.7	92.4	99.1	97.1	95.4	93.5	99.5	11	n estimators 400, max depth None, min samples leaf 1
MLP (pooled features)	97.2	96.5	93.1	99.1	97.7	97.1	94.9	99.2	11	hidden1 64, hidden2 64, dropout 0.1, lr 0.001
1D-CNN (sequence)	97.1	97.0	93.7	99.2	98.0	97.3	95.7	99.0	10	channels 64, kernel 5, layers 3, dropout 0.1, lr 0.001
GRU + attention (sequence)	97.6	95.6	93.9	98.8	98.0	97.3	95.6	98.8	11	hidden 128, layers 1, dropout 0.1, lr 0.001
LSTM + attention (sequence)	97.7	95.6	93.0	99.0	98.4	97.3	93.6	99.1	10	hidden 128, layers 2, dropout 0.2, lr 0.003

Notes. Default: SVM / RF / MLP / CNN / GRU from the B1 run (25 epochs maximum, patience 5, window stride 8); the LSTM default is the deployed model of the cross-validation (same routine, window stride 8). Tuned (B2): 12 configurations drawn at random per model with a fixed seed (with replacement: 10–11 unique per model, the LSTM 10), chosen on the inner validation split of folds 1–3 of seed 1 (mean validation window AUC) and then trained on all 15 fold-models. The search is not nested: those validation traces are test traces of other fold-models, an optimistic bias shared by all six models. Same routine for all neural models (25 epochs maximum, patience 5, about 10 epochs used, AdamW, batch 256, window stride 4); SVM and RF are fitted on random subsamples of 30,000 and 50,000 training windows, the neural models on all windows, so the budget is equal in configurations and epochs, not in data. Like-for-like reference = the tuned LSTM hybrid (same stride and budget); difference of each tuned hybrid to it, dependability / security (pt): CNN +2.2 / -0.1 (n.s.); GRU +2.0 / -0.3 (n.s.); ANN +1.3 / +0.1 (n.s.); RF -0.1 (n.s.) / +0.4 (n.s.); SVM -6.6 / -1.4. The tuned LSTM hybrid vs the deployed hybrid: +0.65 / +0.15 pt (n.s.). Comparisons with the deployed hybrid (post-hoc family) mix window stride 4 with 8 and are listed only in Supplementary Table S4. Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 22 tests. Table names describe architectures; they are not reproductions of the cited papers.

Table S3: Conventional relays on the benchmark data (dependability / security, %): unrestrained dual-slope 87T, harmonic-restrained 87T with textbook settings, and harmonic-restrained 87T tuned on validation traces over the original and over the extended grid (post-hoc; the reference relay of this paper), against the proposed relay. Plant-A CV (3 seeds × 5 folds, common set), OOD (final scoring) and plant-B zero-shot.

Relay	CV		OOD		Plant B	
	dep.	sec.	dep.	sec.	dep.	sec.
87T unrestrained (dual slope, 0.30 pu pickup)	80.3	63.5	89.3	53.3	89.5	62.0
Harmonic 87T, textbook (0.30 pu, k_2 0.15, k_5 0.35)	86.0	99.4	94.0	88.7	94.1	99.0
Harmonic 87T, tuned, original grid	91.2	99.1	n/e	n/e	n/e	n/e
Harmonic 87T, tuned, extended grid[§]	92.8	98.9	97.3	97.3	96.8	99.1
Proposed relay (deployed LSTM hybrid)	93.0	99.0	98.0	96.0	92.7	88.4

Notes. [§] Reference relay of the paper: tuned on validation traces over the extended grid (post-hoc, fixed before its run; OOD and plant B with one setting frozen from plant A): pickup $\{0.1, 0.15, 0.2, 0.3, 0.4\}$ pu × k_2 $\{0.12, 0.15, 0.2, 0.25, 0.3\}$ × k_5 $\{0.15, 0.2, 0.25, 0.35, 0.45\}$, chosen per fold by the same plateau rule; pickup at the lowest grid value in 9 of 15 fold-models (left unresolved: no second extension). Tuned (extended) minus proposed hybrid: dependability -0.16 pt ($p_{\mathrm{H}} = 1.00$; per-seed -0.89, -1.29, +1.69), security -0.02 pt ($p_{\mathrm{H}} = 1.00$): not distinguishable. Original grid: pickup $\{0.2, 0.3, 0.4\}$ pu × k_2 $\{0.12, 0.15, 0.2\}$ × k_5 $\{0.25, 0.35\}$; pickup at the lowest grid value in 15 of 15, k_2 at the highest in 9 of 15, k_5 at the lowest grid value in 15 of 15 fold-models; tuned (original) minus proposed: dependability -1.78 pt ($p_{\mathrm{H}} = 5 \times 10^{-3}$), security +0.12 pt ($p_{\mathrm{H}} = 0.68$). Textbook minus proposed: dependability -6.99 pt ($p_{\mathrm{H}} \leq 2 \times 10^{-5}$), security +0.44 pt ($p_{\mathrm{H}} = 0.16$). Dependability, Internal_HiZ (n=459): hybrid 65.4, tuned harmonic (extended) 69.9, textbook harmonic 51.0; Dependability, other internal faults (n=3258): hybrid 96.8, tuned harmonic (extended) 96.0, textbook harmonic 90.9. Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 4 tests. The original-grid comparison is a separate family of 4. n/e = not evaluated (no new experiments).

Table S4: Instances of the dual-path supervisory logic with interchangeable classifiers (dependability / security, %): plant-A CV (3 seeds × 5 folds, common set, mean over seeds), OOD and plant-B zero-shot where a final model was scored; difference to the validation-tuned harmonic 87T on the extended grid (92.8 / 98.9; post-hoc reference, Holm over 16 tests); and the gated-only ablation (AI-alone path off).

Instance (hybrid mode)	CV		Δ dep. vs tuned	Δ sec. vs tuned	OOD		Plant B	
	dep.	sec.	harmonic 87T (pt)	harmonic 87T (pt)	dep.	sec.	dep.	sec.
LSTM, deployed (closed-loop configuration)	93.0	99.0	$+0.16^{\dagger}$	$+0.02^{\dagger}$	98.0	96.0	92.7	88.4
LSTM, tuned (B2)	93.6	99.1	$+0.81^{\dagger}$	$+0.17^{\dagger}$	n/e	n/e	n/e	n/e
1D-CNN, tuned (B2)	95.7	99.0	$+2.96^{*}$	$+0.08^{\dagger}$	n/e	n/e	n/e	n/e
GRU, tuned (B2)	95.6	98.8	$+2.80^{*}$	$-0.12^{\dagger}$	n/e	n/e	n/e	n/e
MLP, tuned (B2)	94.9	99.2	$+2.13^{*}$	+0.27	n/e	n/e	n/e	n/e
Random forest, tuned (B2)	93.5	99.5	+0.70	$+0.56^{*}$	n/e	n/e	n/e	n/e
Key et al. network (Dis-DNN output)	94.1	99.9	$+1.32^{*}$	$+0.93^{*}$	98.0	95.3	95.5	99.9
Ablation, AI-alone path off (gated only; post-hoc, from saved decisions)								
LSTM, deployed, gated only	88.4	99.4			94.6	98.7	90.5	99.2
Key et al. network, gated only	n/e	n/e			94.6	95.3	94.1	100.0

Notes. * $p_{\mathrm{H}} < 0.05$ (post-hoc family of 16 tests: 12 against the extended-grid tuned harmonic 87T plus its 4 tests); † per-seed differences change sign. Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 16 tests. Deployed LSTM row: from the 4-test family of Table C (deployed minus tuned). 4 of 6 instances are significantly more dependable than the extended-grid tuned harmonic 87T (+1.3 to +3.0 pt; all six +0.7 to +3.0 pt); Key's network, RF significantly more secure; none is significantly less secure. In 3 of 15 CV fold-models of the deployed LSTM the plateau rule switched the AI-alone path off (AI threshold 0.75–0.999), so its CV row mixes dual-path and gated-only fold-models. The B2 instances were evaluated in cross-validation only (n/e = not evaluated). Key's classifier is the Dis-DNN internal-fault probability sampled at the relay's decision instants inside the identical logic. Plant B, deployed hybrid: 87 of 89 false trips first came from the AI-alone path, 2 from the gated path, none from the high-set; Key hybrid: 1 of 757 healthy traces tripped by the AI-alone path, 0 gated, 0 high-set. Gated-only (post-hoc counterfactual from saved decisions, descriptive): on the plant-A CV it loses 168 fault decisions and gains 28 healthy decisions (counted over the three repetitions), -4.52 / +0.47 pt. Key hybrid minus proposed hybrid: dependability +1.16 pt ($p_{\mathrm{H}} = 0.06$), security +0.91 pt ($p_{\mathrm{H}} \leq 2 \times 10^{-4}$); Holm family of 50.

Table S5: Operate time (ms) per method and fault class: median / 90th percentile over correctly tripped fault traces, plant-A CV pooled over 3 seeds × 5 folds, common set, including the like-for-like rows.

Group	Method	All faults	Solid LV / HV	CT saturation	In inrush	SOTF	High impedance	n (all)
Reproduction (native)	Simões	0.4 / 1.3	0.3 / 0.9	0.5 / 0.6	0.3 / 0.6	0.4 / 1.3	0.4 / 7.0	2,590
Reproduction (like-for-like)	Simões, like-for-like	0.3 / 0.6	0.3 / 0.6	0.4 / 0.6	0.5 / 0.6	0.4 / 1.0	0.3 / 0.6	2,856
Reproduction (native)	Ngaopitakkul	5.1 / 65.2	5.1 / 51.4	–	5.2 / 674.1	5.1 / 5.3	5.0 / 5.3	1,030
Reproduction (like-for-like)	Ngaopitakkul, like-for-like	5.0 / 9.6	4.9 / 6.2	–	5.2 / 771.0	5.1 / 515.0	5.0 / 5.3	1,427
Reproduction (native)	Abdusalam	21.1 / 28.5	21.1 / 27.9	23.7 / 40.6	18.7 / 21.2	21.1 / 27.7	19.0 / 21.5	2,839
Reproduction (like-for-like)	Abdusalam, like-for-like	23.3 / 29.4	21.3 / 29.2	23.9 / 31.6	15.7 / 23.0	23.2 / 27.5	26.8 / 29.6	2,945
Reproduction (native)	Afrasiabi	11.2 / 17.7	11.0 / 17.9	11.1 / 13.8	8.3 / 12.2	11.6 / 21.7	16.1 / 24.1	3,223
Reproduction (like-for-like)	Afrasiabi, like-for-like	22.3 / 28.9	22.2 / 29.1	22.3 / 25.4	18.6 / 21.8	22.7 / 28.4	27.4 / 35.3	3,244
Reproduction (native)	Key	18.1 / 19.7	18.1 / 19.4	18.4 / 20.0	15.9 / 19.7	19.1 / 47.5	18.1 / 21.2	3,605
Reproduction (like-for-like)	Key, like-for-like	30.0 / 32.5	30.0 / 31.2	30.6 / 31.9	30.0 / 41.1	31.2 / 113.1	30.0 / 34.1	3,614
Reproduction (native)	Alhamd	15.2 / 20.2	15.0 / 20.0	16.7 / 20.6	14.2 / 19.5	19.4 / 21.9	12.8 / 19.8	3,588
Reproduction (like-for-like)	Alhamd, like-for-like	26.2 / 32.5	23.8 / 31.7	29.6 / 32.9	27.3 / 32.9	31.2 / 33.9	26.4 / 32.7	3,640
Conventional relay	87T unrestrained	1.2 / 7.5	1.2 / 8.1	1.2 / 3.8	–	1.9 / 6.2	6.2 / 14.4	2,985
	Harmonic 87T, textbook (single pass)	16.9 / 20.0	16.2 / 19.4	17.5 / 20.0	17.2 / 377.8	16.9 / 87.3	17.2 / 18.9	1,065
Dual-path instance	LSTM, deployed	16.2 / 36.2	15.6 / 31.2	16.2 / 27.5	28.1 / 45.6	21.2 / 163.4	25.0 / 45.7	3,455
	Key's network	30.0 / 33.8	30.0 / 31.9	30.6 / 31.9	30.0 / 40.6	31.2 / 98.9	30.0 / 41.2	3,498
	1D-CNN	16.2 / 31.9	15.0 / 26.0	16.2 / 20.0	23.1 / 38.1	19.4 / 151.0	19.4 / 39.9	3,559
	GRU	15.0 / 29.4	14.4 / 23.1	15.0 / 21.2	20.6 / 38.8	18.4 / 46.6	19.4 / 43.0	3,553
	MLP	16.2 / 38.8	15.6 / 33.1	16.2 / 19.4	35.0 / 43.1	35.6 / 160.5	20.6 / 51.2	3,528
	Random forest	17.5 / 50.0	15.6 / 46.2	19.4 / 35.5	37.5 / 48.9	30.0 / 143.5	31.9 / 68.7	3,475
	LSTM, tuned	15.6 / 30.0	15.0 / 19.4	15.6 / 20.0	21.9 / 41.2	16.9 / 45.8	16.2 / 39.4	3,479
	SVM	17.5 / 37.5	16.2 / 35.0	16.9 / 35.6	35.6 / 47.4	35.6 / 40.4	31.9 / 38.1	3,234
Ablation	LSTM, deployed, gated only	16.2 / 33.1	15.6 / 26.9	16.2 / 27.5	28.1 / 45.6	21.2 / 163.4	20.6 / 36.2	3,287
	LSTM alone	15.6 / 31.9	15.0 / 22.5	15.6 / 20.0	23.8 / 41.4	18.8 / 112.9	18.1 / 39.4	3,630

Notes. Operate time = trip time minus the applied fault time (verified on the waveform by the generator check); counts are (seed, fold, trace) decisions, so each trace appears up to three times; descriptive, no test. Reproductions run at their own sampling rates and decide once per detector firing (Simões, Ngaopitakkul) or per window; the harmonic 87T row is a single pass over all traces with textbook settings (no operate times were saved for the tuned settings). Evolving faults are counted only in 'All faults'. '–' = no correctly tripped trace in the class.

S2 Reproduced methods

Specifications. Each method was specified in writing before its code was run: the paper's own choices, the details the paper leaves open ("fills", each with a reason), the sampling rate, the decision rule of the native mode and a sanity check. Later amendments are appended to the same files with their date and reason. The files are:

- `documents/baselines/abdusalam2022_dwt_omp_ga_spec.md`: Spec: Abdusalam, Ibrahim, Anayi and Packianather 2022 (DWT + OMP features, bees / genetic feature selection, KNN / SVM / ANN)
- `documents/baselines/afrasiabi2020_acnn_spec.md`: Spec: Afrasiabi, Afrasiabi, Parang and Mohammadi 2020 (accelerated CNN on half-cycle differential currents)
- `documents/baselines/alhamd2024_wavelet_lstm_spec.md`: Spec: Alhamd, Saniei, Seifossadat and Mashhour 2024 (improved wavelet analysis + bidirectional LSTM)
- `documents/baselines/hajipour2015_ct_comp_spec.md`: Spec: Hajipour, Vakilian and Sanaye-Pasand 2015 (CT saturation compensation) - first stage of Afrasiabi et al. 2020
- `documents/baselines/key2025_comdnn_disdnn_spec.md`: Spec: Key, Leap, Yoon, Lee and Nam 2025 (Com-DNN CT-saturation compensation + Dis-DNN stacked denoising autoencoders)
- `documents/baselines/ngaopitakkul2006_dwt_bpnn_spec.md`: Spec: Ngaopitakkul and Kunakorn 2006 (db4 DWT + back-propagation network)
- `documents/baselines/simoes2021_rtbswt_svm_spec.md`: Spec: Simões et al. 2021 (RT-BSWT + two SVMs)

Run-to-run variation. Training of the neural reproductions of Key and Afrasiabi is not bit-reproducible. Two runs on identical folds differ on 74 (Key) and 367 (Afrasiabi) of 9,906 (seed, fold, trace) decisions; the paper uses the second run for every row of Table A. The comparison of Afrasiabi with and without the CT compensation stage in the notes of Table S1 mixes the two runs and is of the size of this variation; it supports only the statement that the stage is nearly inactive on these traces.

Operating conditions that disadvantage a method. The lumped plants carry little energy above 1.6 kHz (no winding capacitances), which limits the high-frequency detectors of Simões et al. and Ngaopitakkul and Kunakorn; the detector of the latter passes its sanity check on noise-free currents but not on CT secondaries with measurement noise. The feature stage of Abdusalam et al. needs 609 ms per window against a 2.5 ms budget (not real-time on the development PC). These conditions are reported, not corrected.

S3 Paired tests

Every paired comparison behind a significance mark in the paper. The marks use the trace-clustered test; the pooled McNemar values, which count each trace up to 3 times, are shown only to make the change of test unit transparent.

Table S6: Reproductions vs the proposed relay and vs the LSTM alone, native and like-for-like. Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 48 tests. Pooled = exact McNemar on (seed, fold, trace) decisions, Holm in the same family (each trace counted up to three times; shown for transparency, not used for the marks).

Comparison (difference = first minus second, pt)	diff.	p_{H} pooled	p_{H} clustered
Proposed hybrid vs Simões [native] dependability	-23.27	3×10^{-169}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs Simões [native] security	-6.75	8×10^{-80}	$\leq 2 \times 10^{-4}$*
LSTM alone vs Simões [native] dependability	-27.98	0×10^{0}	$\leq 2 \times 10^{-4}$*
LSTM alone vs Simões [native] security	-3.37	4×10^{-14}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs afrasiabi [native] dependability	-6.24	1×10^{-45}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs afrasiabi [native] security	-10.32	1×10^{-151}	$\leq 2 \times 10^{-4}$*
LSTM alone vs afrasiabi [native] dependability	-10.95	2×10^{-95}	$\leq 2 \times 10^{-4}$*
LSTM alone vs afrasiabi [native] security	-6.94	1×10^{-47}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs key [native] dependability	+4.04	4×10^{-15}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs key [native] security	-23.67	0×10^{0}	$\leq 2 \times 10^{-4}$*
LSTM alone vs key [native] dependability	-0.67	0.25	0.67
LSTM alone vs key [native] security	-20.29	1×10^{-241}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs alhamd [native] dependability	+3.58	6×10^{-12}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs alhamd [native] security	-16.04	1×10^{-250}	$\leq 2 \times 10^{-4}$*
LSTM alone vs alhamd [native] dependability	-1.13	9×10^{-3}	0.09
LSTM alone vs alhamd [native] security	-12.66	9×10^{-131}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs abdusalam_ba_svm [native] dependability	-16.57	1×10^{-172}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs abdusalam_ba_svm [native] security	+1.00	3×10^{-14}	$\leq 2 \times 10^{-4}$*
LSTM alone vs abdusalam_ba_svm [native] dependability	-21.28	1×10^{-234}	$\leq 2 \times 10^{-4}$*
LSTM alone vs abdusalam_ba_svm [native] security	+4.38	3×10^{-72}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs ngaopitakkul [native] dependability	-65.24	0×10^{0}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs ngaopitakkul [native] security	+0.30	0.29	0.67
LSTM alone vs ngaopitakkul [native] dependability	-69.95	0×10^{0}	$\leq 2 \times 10^{-4}$*
LSTM alone vs ngaopitakkul [native] security	+3.69	6×10^{-38}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs Simões [modeb] dependability	-16.12	4×10^{-88}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs Simões [modeb] security	-26.87	0×10^{0}	$\leq 2 \times 10^{-4}$*
LSTM alone vs Simões [modeb] dependability	-20.82	2×10^{-183}	$\leq 2 \times 10^{-4}$*
LSTM alone vs Simões [modeb] security	-23.49	1×10^{-278}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs afrasiabi [modeb] dependability	-5.68	5×10^{-47}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs afrasiabi [modeb] security	+0.69	6×10^{-6}	6×10^{-4}*
LSTM alone vs afrasiabi [modeb] dependability	-10.38	9×10^{-100}	$\leq 2 \times 10^{-4}$*
LSTM alone vs afrasiabi [modeb] security	+4.08	9×10^{-58}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs key [modeb] dependability	+4.28	4×10^{-24}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs key [modeb] security	+0.61	1×10^{-3}	0.02*

Comparison	diff.	p_{H} pooled	p_{H} clustered
LSTM alone vs key [modeb] dependability	-0.43	0.30	0.67
LSTM alone vs key [modeb] security	+3.99	2×10^{-49}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs alhamd [modeb] dependability	+4.98	6×10^{-36}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs alhamd [modeb] security	-0.78	1×10^{-3}	0.02*
LSTM alone vs alhamd [modeb] dependability	+0.27	0.36	0.67
LSTM alone vs alhamd [modeb] security	+2.61	4×10^{-19}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs abdusalam_ba_svm [modeb] dependability	-13.72	1×10^{-132}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs abdusalam_ba_svm [modeb] security	-6.04	5×10^{-70}	$\leq 2 \times 10^{-4}$*
LSTM alone vs abdusalam_ba_svm [modeb] dependability	-18.43	4×10^{-193}	$\leq 2 \times 10^{-4}$*
LSTM alone vs abdusalam_ba_svm [modeb] security	-2.66	6×10^{-9}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs ngaopitakkul [modeb] dependability	-54.56	0×10^{0}	$\leq 2 \times 10^{-4}$*
Proposed hybrid vs ngaopitakkul [modeb] security	-1.47	7×10^{-9}	2×10^{-4}*
LSTM alone vs ngaopitakkul [modeb] dependability	-59.27	0×10^{0}	$\leq 2 \times 10^{-4}$*
LSTM alone vs ngaopitakkul [modeb] security	+1.91	2×10^{-7}	3×10^{-4}*

Table S7: Same-feature hybrids (B2): vs the tuned LSTM hybrid (planned) and vs the deployed hybrid (post-hoc; window stride 4 vs 8). Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 22 tests. Pooled = exact McNemar on (seed, fold, trace) decisions, Holm in the same family (each trace counted up to three times; shown for transparency, not used for the marks).

Comparison (difference = first minus second, pt)	diff.	p_{H} pooled	p_{H} clustered
Wavelet-SVM hybrid vs tuned LSTM hybrid dependability	-6.59	2×10^{-57}	$\leq 1 \times 10^{-4}$*
Wavelet-SVM hybrid vs tuned LSTM hybrid security	-1.37	1×10^{-8}	1×10^{-4}*
Wavelet-RF hybrid vs tuned LSTM hybrid dependability	-0.11	1.00	1.00
Wavelet-RF hybrid vs tuned LSTM hybrid security	+0.39	0.10	0.49
Wavelet-ANN hybrid vs tuned LSTM hybrid dependability	+1.32	1×10^{-3}	0.02*
Wavelet-ANN hybrid vs tuned LSTM hybrid security	+0.10	1.00	1.00
1D-CNN hybrid vs tuned LSTM hybrid dependability	+2.15	8×10^{-11}	$\leq 1 \times 10^{-4}$*
1D-CNN hybrid vs tuned LSTM hybrid security	-0.08	1.00	1.00
Wavelet-GRU hybrid vs tuned LSTM hybrid dependability	+1.99	3×10^{-8}	$\leq 1 \times 10^{-4}$*

Comparison	diff.	p_{H} pooled	p_{H} clustered
Wavelet-GRU hybrid vs tuned LSTM hybrid security	-0.29	0.36	0.72
tuned LSTM hybrid vs deployed hybrid dependability	+0.65	0.79	0.72
tuned LSTM hybrid vs deployed hybrid security	+0.15	1.00	1.00
Wavelet-SVM hybrid vs deployed hybrid dependability	-5.95	6×10^{-57}	$\leq 1 \times 10^{-4}$*
Wavelet-SVM hybrid vs deployed hybrid security	-1.22	3×10^{-7}	2×10^{-4}*
Wavelet-RF hybrid vs deployed hybrid dependability	+0.54	0.90	1.00
Wavelet-RF hybrid vs deployed hybrid security	+0.54	5×10^{-3}	0.03*
Wavelet-ANN hybrid vs deployed hybrid dependability	+1.96	3×10^{-8}	$\leq 1 \times 10^{-4}$*
Wavelet-ANN hybrid vs deployed hybrid security	+0.25	0.90	1.00
1D-CNN hybrid vs deployed hybrid dependability	+2.80	1×10^{-17}	$\leq 1 \times 10^{-4}$*
1D-CNN hybrid vs deployed hybrid security	+0.07	1.00	1.00
Wavelet-GRU hybrid vs deployed hybrid dependability	+2.64	5×10^{-15}	$\leq 1 \times 10^{-4}$*
Wavelet-GRU hybrid vs deployed hybrid security	-0.14	1.00	1.00

Table S8: Instances of the logic vs the validation-tuned harmonic 87T (post-hoc). Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 16 tests. Pooled = exact McNemar on (seed, fold, trace) decisions, Holm in the same family (each trace counted up to three times; shown for transparency, not used for the marks).

Comparison (difference = first minus second, pt)	diff.	p_{H} pooled	p_{H} clustered
1D-CNN hybrid vs tuned harmonic dependability	+4.57	3×10^{-37}	$\leq 8 \times 10^{-5}$*
1D-CNN hybrid vs tuned harmonic security	-0.05	1.00	1.00
Wavelet-GRU hybrid vs tuned harmonic dependability	+4.41	3×10^{-31}	$\leq 8 \times 10^{-5}$*
Wavelet-GRU hybrid vs tuned harmonic security	-0.25	1.00	1.00
Wavelet-ANN hybrid vs tuned harmonic dependability	+3.74	2×10^{-27}	$\leq 8 \times 10^{-5}$*
Wavelet-ANN hybrid vs tuned harmonic security	+0.14	1.00	1.00
Key hybrid vs tuned harmonic dependability	+2.93	1×10^{-18}	$\leq 8 \times 10^{-5}$*
Key hybrid vs tuned harmonic security	+0.80	1×10^{-8}	1×10^{-3}*
Wavelet-RF hybrid vs tuned harmonic dependability	+2.31	4×10^{-13}	1×10^{-4}*
Wavelet-RF hybrid vs tuned harmonic security	+0.42	0.01	0.19
Proposed LSTM hybrid vs tuned harmonic dependability	+2.42	1×10^{-7}	1×10^{-3}*
Proposed LSTM hybrid vs tuned harmonic security	+0.03	1.00	1.00

Table S9: Harmonic 87T (textbook and tuned) vs the proposed relay. Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 4 tests. Pooled = exact McNemar on (seed, fold, trace) decisions, Holm in the same family (each trace counted up to three times; shown for transparency, not used for the marks).

Comparison (difference = first minus second, pt)	diff.	p_{H} pooled	p_{H} clustered
textbook harmonic vs proposed hybrid dependability	-6.99	6×10^{-69}	$\leq 2 \times 10^{-5}$*
textbook harmonic vs proposed hybrid security	+0.44	0.01	0.11
tuned harmonic vs proposed hybrid dependability	-1.78	2×10^{-5}	5×10^{-3}*
tuned harmonic vs proposed hybrid security	+0.12	0.55	0.68

Table S10: Key's network inside the logic vs the deployed LSTM instance. Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 50 tests. Pooled = exact McNemar on (seed, fold, trace) decisions, Holm in the same family (each trace counted up to three times; shown for transparency, not used for the marks).

Comparison (difference = first minus second, pt)	diff.	p_{H} pooled	p_{H} clustered
Key hybrid vs proposed hybrid dependability	+1.16	1×10^{-2}	0.06
Key hybrid vs proposed hybrid security	+0.91	2×10^{-10}	$\leq 2 \times 10^{-4}$*

Table S11: Harmonic 87T on the extended grid (post-hoc reference) and textbook settings vs the proposed relay. Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 4 tests. Pooled = exact McNemar on (seed, fold, trace) decisions, Holm in the same family (each trace counted up to three times; shown for transparency, not used for the marks).

Comparison (difference = first minus second, pt)	diff.	p_{H} pooled	p_{H} clustered
textbook harmonic vs proposed hybrid dependability	-6.99	6×10^{-69}	$\leq 2 \times 10^{-5}$*
textbook harmonic vs proposed hybrid security	+0.44	0.02	0.16
tuned harmonic vs proposed hybrid dependability	-0.16	1.00	1.00
tuned harmonic vs proposed hybrid security	-0.02	1.00	1.00

Table S12: Instances of the logic vs the harmonic 87T tuned on the extended grid (post-hoc reference). Significance: trace-clustered sign-flip permutation test (test unit = trace; per-trace difference averaged over the three CV repetitions; 200,000 draws), Holm-adjusted within a family of 16 tests. Pooled = exact McNemar on (seed, fold, trace) decisions, Holm in the same family (each trace counted up to three times; shown for transparency, not used for the marks).

Comparison (difference = first minus second, pt)	diff.	p_{H} pooled	p_{H} clustered
1D-CNN hybrid vs tuned harmonic dependability	+2.96	2×10^{-16}	$\leq 8 \times 10^{-5}$*
1D-CNN hybrid vs tuned harmonic security	+0.08	1.00	1.00
Wavelet-GRU hybrid vs tuned harmonic dependability	+2.80	8×10^{-14}	$\leq 8 \times 10^{-5}$*
Wavelet-GRU hybrid vs tuned harmonic security	-0.12	1.00	1.00
Wavelet-ANN hybrid vs tuned harmonic dependability	+2.13	2×10^{-9}	4×10^{-4}*
Wavelet-ANN hybrid vs tuned harmonic security	+0.27	0.75	1.00
Key hybrid vs tuned harmonic dependability	+1.32	2×10^{-4}	0.03*
Key hybrid vs tuned harmonic security	+0.93	1×10^{-10}	3×10^{-4}*
Wavelet-RF hybrid vs tuned harmonic dependability	+0.70	0.34	1.00
Wavelet-RF hybrid vs tuned harmonic security	+0.56	3×10^{-4}	0.03*
Proposed LSTM hybrid vs tuned harmonic dependability	+0.81	0.48	1.00
Proposed LSTM hybrid vs tuned harmonic security	+0.17	1.00	1.00

S4 Order of decisions and analyses

Decision rules were fixed in a dated plan before each analysis (`BENCHMARK_FAIRNESS_PLAN.md`, with the git history of the repository as the time record; progress log `IMPLEMENTATION_PLAN.md`). The hypotheses and decision tables for the reproductions, the statistics and the conventional baselines were committed before any reproduction was evaluated, and were not edited afterwards. The following analyses were added after earlier results had been seen and are labelled post-hoc in the paper: the comparison of every instance with the tuned harmonic 87T and the range-only wording for it; the B2 comparisons with the deployed instance; the gated-only ablation; the extension of the harmonic grid of the tuned 87T to 125 settings, with its decision rules (restate the claim if the tuned relay is not significantly less dependable than the deployed instance; report a new boundary as unresolved, no second extension) fixed before its run, and every comparison with the relay tuned on it, including its frozen-setting scoring on the shifted-range set and on plant B; the extension of the claim to the like-for-like mode; the re-measurement of the latency of Alhamd's trained configurations; and the larger training budget of Alhamd's reproduction (both budgets are reported). Some review sections of the plan were committed together after the results they discuss; their timing rests on the dated log.

S5 Audit

An independent audit (`audit/` in the repository) recomputed every reported test from the per-trace files, re-tested them with the trace-clustered permutation test used in the paper, checked the code for leakage

(grouping, scaling, threshold selection, use of ground truth at test time) and verified every reference against a publisher record. Its findings that affect the paper are taken in: the test unit, the window stride of the deployed instance (8 against 4 in B2), the non-nested B2 search, the subsampling of SVM and random forest, the label definition from the applied fault time, and the classifier-alone path being switched off by the plateau rule in 3 of 15 CV fold-models.